

August 1, 2026

Editor-in-Chief  
*Performance Evaluation*

**Re:** Submission of “Nominal Budgets and Realized Resources in Closed-API Large-Language-Model Inference: A Performance Evaluation Protocol”

Dear Editor-in-Chief,

I respectfully submit the enclosed manuscript, “Nominal Budgets and Realized Resources in Closed-API Large-Language-Model Inference: A Performance Evaluation Protocol,” for consideration for publication in *Performance Evaluation*.

Budgeted closed-API large language model (LLM) evaluations are often summarized by accuracy under a shared nominal inference budget. Equal nominal budgets need not imply equal realized resource use, and they can conflate candidate discovery with final-answer selection. Closed commercial APIs intensify the measurement problem: providers differ in operational behavior, telemetry may be incomplete, and attribution and reproducibility are correspondingly difficult. Without an auditable accounting contract, budget-matched comparisons can be mistaken for resource-matched ones, and selector gains for stronger discovery.

The manuscript contributes a unified, auditable performance-evaluation protocol for closed-API, budgeted inference. The protocol separates nominal budget from realized resources, separates candidate discovery from final-answer selection, evaluates identical-pool selector controls, labels provider transfer for failure-mined rules, and reports incomplete telemetry and protocol-blocked outcomes rather than silently dropping or imputing them. Named generators, adapters, and selectors are measurement objects used to exercise the protocol, not a general ranking of reasoning systems.

Empirically, the protocol is exercised on four providers and four datasets, yielding 15 completed provider-by-dataset cells with 3,394 paired examples and one explicitly retained protocol-blocked cell (Fireworks×GPQA-Diamond). On the shared four-generator pool, identical-pool plurality (Pooled-4) exceeds the failure-trace gated selector (FTA) overall (66.53% versus 65.00%; McNemar  $p=0.00027$ ). Transfer failures concentrate primarily in the dominant FIX-2 gate, especially in Azure mathematics cells. Reconstructing successful-completion counts changes interpretation of nominally matched methods: Frontier successful completions rise from 2.78 to 5.38 of nominal budget  $B=6$ , while token and dollar fields remain lower bounds.

The manuscript fits *Performance Evaluation* as a measurement-methodology contribution concerning workload and resource semantics under closed-service benchmarking, experimental attribution between discovery and selection, and reproducibility practice when telemetry is incomplete. Its claim is that defensible closed-API inference-performance conclusions require an explicit resource layer and attribution controls. The manuscript does not claim a universally superior reasoning algorithm; official reproduction of L1, s1, or TALE; a complete cost-efficiency or Pareto analysis; or full regeneration of proprietary API outputs.

This manuscript is original, has not been published elsewhere, and is not under consideration elsewhere. I am the sole author and approve this submission. Funding and competing-interest disclosures are included with the manuscript. Evaluation code and frozen aggregate audit artifacts supporting verification of the reported aggregates are publicly available at <https://github.com/SoroushVahidi/frontier-allocation-for-budgeted-llm-inference>.

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Thank you for your consideration.

Sincerely,

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